

Ibn Tofail University

Analysis IV

Problem Set 2

Exercise 1:

Study the nature of convergence of each of the following sequences of functions:

1. $f_n(x) = \frac{1}{1+n(nx-1)^2}$, for $x \in I = [0, 1]$.

2. $f_n(x) = \begin{cases} (n-1)x & \text{if } x \in [0, \frac{1}{n}] , \\ 1-x & \text{if } x \in [\frac{1}{n}, 1] . \end{cases}$

3. $f_n(x) = e^{-nx} \cos x^2$, for $x > 0$.

Correction

1. Pointwise Convergence on $I = [0, 1]$:

- For $x = 0$, $f_n(0) = \frac{1}{1+n(n \cdot 0 - 1)^2} = \frac{1}{1+n}$. Thus, $\lim_{n \rightarrow +\infty} f_n(0) = 0$.
- For $x > 0$ (and $x \in [0, 1]$), the term $n(nx-1)^2$ tends to $+\infty$ as $n \rightarrow +\infty$.
 - If $x \neq \frac{1}{n}$, then $f_n(x) = \frac{1}{1+n(nx-1)^2} \rightarrow 0$ as $n \rightarrow +\infty$.
 - If $x = \frac{1}{n}$ (for large n), $f_n(\frac{1}{n}) = \frac{1}{1+n(\frac{1}{n}-1)^2} = 1$. This specific point $\frac{1}{n}$ approaches 0, but the limit x is fixed, so $f_n(x) \rightarrow 0$ for a fixed $x > 0$.

The sequence (f_n) converges pointwise to the function $f(x) = 0$ on $I = [0, 1]$.

Uniform Convergence on $I = [0, 1]$: We consider the norm $|f_n - f| = \sup_{x \in [0, 1]} |f_n(x) - 0| = \sup_{x \in [0, 1]} \frac{1}{1+n(nx-1)^2}$. The maximum value of $f_n(x)$ occurs when the denominator is minimized, i.e., when $(nx-1)^2 = 0$, or $x = \frac{1}{n}$. Since $n \geq 1$, $\frac{1}{n} \in [0, 1]$, so $x = \frac{1}{n}$ is in the interval.

$$\sup_{x \in [0, 1]} f_n(x) = f_n\left(\frac{1}{n}\right) = \frac{1}{1+n(n\frac{1}{n}-1)^2} = \frac{1}{1+0} = 1.$$

Since $\lim_{n \rightarrow +\infty} \sup_{x \in [0, 1]} |f_n(x) - f(x)| = \lim_{n \rightarrow +\infty} 1 = 1 \neq 0$, the convergence is **not uniform** on $I = [0, 1]$.

2. Pointwise Convergence on $I = [0, 1]$: We determine the limit function $f(x) = \lim_{n \rightarrow +\infty} f_n(x)$.

- For $x = 0$, $f_n(0) = (n - 1) \cdot 0 = 0$. Thus, $\lim_{n \rightarrow +\infty} f_n(0) = 0$.
- For $x \in (0, 1]$, for n large enough such that $\frac{1}{n} < x$ (i.e., $n > \frac{1}{x}$), $f_n(x) = 1 - x$. Thus, $\lim_{n \rightarrow +\infty} f_n(x) = 1 - x$.

The sequence (f_n) converges pointwise to $f(x) = 1 - x$ on $I = [0, 1]$. **Uniform Convergence on $I = [0, 1]$:** The functions $f_n(x)$ are all continuous on $I = [0, 1]$. However, the limit function is $f(x) = 1 - x$, which is continuous on $I = [0, 1]$. The Continuity Criterion does not help determine the convergence type in this case. The difference $|f_n(x) - f(x)|$ is:

$$|f_n(x) - f(x)| = \begin{cases} |(n-1)x - (1-x)| = |nx - 1| & \text{if } x \in [0, \frac{1}{n}] , \\ |1-x - (1-x)| = 0 & \text{if } x \in [\frac{1}{n}, 1] . \end{cases}$$

We compute the norm: $|f_n - f| = \sup_{x \in [0, 1]} |f_n(x) - f(x)| = \sup_{x \in [0, \frac{1}{n}]} |nx - 1|$. The function $g_n(x) = |nx - 1|$ is $\begin{cases} 1 - nx & \text{if } x \in [0, 1/n] . \end{cases}$ Since $g_n(x)$ is decreasing on $[0, 1/n]$, the maximum occurs at $x = 0$:

$$\sup_{x \in [0, 1]} |f_n(x) - f(x)| = g_n(0) = |n \cdot 0 - 1| = 1.$$

Since $\lim_{n \rightarrow +\infty} \sup_{x \in [0, 1]} |f_n(x) - f(x)| = \lim_{n \rightarrow +\infty} 1 = 1 \neq 0$, the convergence is **not uniform** on $I = [0, 1]$.

3. **Pointwise Convergence on $I = (0, +\infty)$:** For any fixed $x > 0$, we consider the limit $f(x) = \lim_{n \rightarrow +\infty} f_n(x) = \lim_{n \rightarrow +\infty} e^{-nx} \cos x^2$. Since $x > 0$, $\lim_{n \rightarrow +\infty} e^{-nx} = 0$. Since $\cos x^2$ is bounded ($|\cos x^2| \leq 1$), the product converges to 0.

$$\lim_{n \rightarrow +\infty} f_n(x) = 0 \cdot \cos x^2 = 0.$$

The sequence (f_n) converges pointwise to the function $f(x) = 0$ on $I = (0, +\infty)$. **Uniform Convergence on $I = (0, +\infty)$:** We consider the norm $|f_n - f| = \sup_{x > 0} |e^{-nx} \cos x^2 - 0|$.

$$|f_n - f| = \sup_{x > 0} |e^{-nx} \cos x^2| \geq \sup_{x > 0} e^{-nx} |\cos x^2|.$$

The maximum value of e^{-nx} occurs as $x \rightarrow 0^+$, where $\lim_{x \rightarrow 0^+} e^{-nx} = 1$. Near $x = 0$, $\cos x^2 \approx 1$. The supremum is at least 1. Since e^{-nx} decreases as x increases, and $\cos x^2$ oscillates, the supremum is hard to find exactly, but we can check the limit as $x \rightarrow 0^+$. For any $\epsilon < 1$ (e.g., $\epsilon = 1/2$), we need to find η such that for $n \geq \eta$, $|f_n(x)| \leq \epsilon$ for all $x > 0$. If we pick $x_n = \frac{1}{n} \in (0, +\infty)$, then $f_n(x_n) = e^{-n(1/n)} \cos(1/n^2) = e^{-1} \cos(1/n^2)$.

$$\lim_{n \rightarrow +\infty} |f_n - f| \geq \lim_{n \rightarrow +\infty} |f_n(x_n)| = \lim_{n \rightarrow +\infty} e^{-1} \cos(1/n^2) = e^{-1} \cos(0) = e^{-1} \neq 0.$$

Since $e^{-1} \approx 0.368 \neq 0$, the convergence is **not uniform** on $I = (0, +\infty)$.

Exercise 2:

Consider the sequence of functions $f_n : [-1, 1] \rightarrow \mathbb{R}$, defined by

$$x \mapsto f_n(x) = e^{-nx^2} \sin nx + \sqrt{1-x^2}, \quad n \in \mathbb{N}.$$

1. Show that the sequence of functions (f_n) converges pointwise on $I = [-1, 1]$ to a function f , which you will determine.
2. Show that (f_n) converges uniformly to f on $[\omega, 1]$, where ω is a positive constant.
3. Show that (f_n) does not converge uniformly to f on $[0, 1]$.

Correction

1. **Pointwise Convergence on $I = [-1, 1]$:** The limit of $f_n(x)$ is $f(x) = \lim_{n \rightarrow +\infty} (e^{-nx^2} \sin nx + \sqrt{1-x^2})$.

- For $x = 0$: $f_n(0) = e^0 \sin(0) + \sqrt{1-0} = 0 + 1 = 1$. Thus, $\lim_{n \rightarrow +\infty} f_n(0) = 1$.
- For $x \in [-1, 1] \setminus \{0\}$ (i.e., $x \neq 0$): $x^2 > 0$.

$$\lim_{n \rightarrow +\infty} e^{-nx^2} = 0.$$

Since $|\sin nx| \leq 1$, the product $e^{-nx^2} \sin nx$ converges to 0.

$$\lim_{n \rightarrow +\infty} f_n(x) = 0 + \sqrt{1-x^2} = \sqrt{1-x^2}.$$

The sequence (f_n) converges pointwise to the function $f(x)$ on $I = [-1, 1]$, defined as:

$$f(x) = \sqrt{1-x^2}$$

2. **Uniform Convergence on $[\omega, 1]$ with $\omega > 0$:** We study $|f_n - f| = \sup_{x \in [\omega, 1]} |f_n(x) - f(x)|$.

$$|f_n(x) - f(x)| = |e^{-nx^2} \sin nx + \sqrt{1-x^2} - \sqrt{1-x^2}| = |e^{-nx^2} \sin nx|.$$

For $x \in [\omega, 1]$, we have $x^2 \geq \omega^2 > 0$. Since $|\sin nx| \leq 1$, we can bound the difference:

$$|f_n(x) - f(x)| = |e^{-nx^2} \sin nx| \leq e^{-nx^2} \leq e^{-n\omega^2}.$$

Thus, the uniform norm is bounded by:

$$|f_n - f| = \sup_{x \in [\omega, 1]} |f_n(x) - f(x)| \leq e^{-n\omega^2}.$$

Since $\omega > 0$, $\omega^2 > 0$, and $\lim_{n \rightarrow +\infty} e^{-n\omega^2} = 0$. Therefore, $\lim_{n \rightarrow +\infty} |f_n - f| = 0$, and the convergence is **uniform** on $[\omega, 1]$.

3. **Lack of Uniform Convergence on $[0, 1]$:** We study the uniform norm on $[0, 1]$: $|f_n - f| = \sup_{x \in [0, 1]} |e^{-nx^2} \sin nx|$. The maximum of $|e^{-nx^2} \sin nx|$ is expected to occur near $x = 0$, where e^{-nx^2} is close to 1. Consider the sequence of points $x_n = \frac{\pi}{2n} \in [0, 1]$. For sufficiently large n , x_n is close to 0.

$$|f_n(x_n) - f(x_n)| = \left| e^{-n(\frac{\pi}{2n})^2} \sin \left(n \frac{\pi}{2n} \right) \right| = \left| e^{-\frac{\pi^2}{4n}} \sin \left(\frac{\pi}{2} \right) \right| = e^{-\frac{\pi^2}{4n}}.$$

The limit of this difference is:

$$\lim_{n \rightarrow +\infty} |f_n(x_n) - f(x_n)| = \lim_{n \rightarrow +\infty} e^{-\frac{\pi^2}{4n}} = e^0 = 1.$$

Since $|f_n - f| = \sup_{x \in [0, 1]} |f_n(x) - f(x)| \geq |f_n(x_n) - f(x_n)|$, we have:

$$\lim_{n \rightarrow +\infty} |f_n - f| \geq 1 \neq 0.$$

Therefore, the convergence is **not uniform** on $[0, 1]$.

Exercise 3:

Determine the domain of convergence D of each series:

1. $\sum_{n>0} \frac{\cos nx}{n!}$.
2. $\sum_{n>1} \frac{1}{(z-n)^2}$.
3. $\sum_{n>1} \left| \sin \frac{x}{2^n} - \tan \frac{x}{2^n} \right|^{\frac{1}{2}}$.
4. $\sum_{n>1} \frac{n!(x-3)^{2n}}{n^{n+1}}$.

Correction

1. Let $u_n(x) = \frac{\cos nx}{n!}$. Since $|\cos nx| \leq 1$ for all $x \in \mathbb{R}$, we have:

$$|u_n(x)| = \left| \frac{\cos nx}{n!} \right| \leq \frac{1}{n!}.$$

The series $\sum_{n>0} \frac{1}{n!}$ is a convergent series (it converges to $e - 1$). By the ****Weierstrass M-test**** (comparison with a convergent series of positive terms, which implies uniform convergence, and thus, pointwise convergence), the series $\sum u_n(x)$ is ****absolutely and uniformly convergent**** on $\mathbb{R}$. The domain of convergence is $D = \mathbb{R}$. —

2. This is a series of complex functions. We assume $z \in \mathbb{C}$. The series is defined when $z - n \neq 0$, i.e., $z \neq n$ for all $n \in \mathbb{N}, n \geq 2$. The domain where the series is defined is $D_0 = \mathbb{C} \setminus \{2, 3, 4, \dots\}$. Let $u_n(z) = \frac{1}{(z-n)^2}$. For a fixed $z \in D_0$, let N be an integer such that $|z - n| \geq 1$ for all $n \geq N$ (this is always possible since $\lim_{n \rightarrow +\infty} |z - n| = +\infty$). For $n \geq N$, we have:

$$|u_n(z)| = \left| \frac{1}{(z-n)^2} \right| = \frac{1}{|z-n|^2}.$$

Since $|z-n| \rightarrow \infty$ as $n \rightarrow \infty$, for n large enough, $|z-n|^2 \approx n^2$. More formally, for $n > |z| + 1$, we have $|z-n| \geq |n-|z|| \geq n-|z| > n-(n-1) = 1$. Using the limit comparison test, $\lim_{n \rightarrow +\infty} \frac{1/|z-n|^2}{1/n^2} = 1 \neq 0, \infty$. Since $\sum \frac{1}{n^2}$ converges (p-series with $p = 2 > 1$), the series $\sum |u_n(z)|$ converges by comparison. The series $\sum_{n>1} \frac{1}{(z-n)^2}$ is ****absolutely convergent**** on $D = \mathbb{C} \setminus \{n \in \mathbb{N} \mid n \geq 2\}$.
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3. Let $u_n(x)$ be the general term. For $x \in \mathbb{R}$, as $n \rightarrow +\infty$, $\frac{x}{2^n} \rightarrow 0$. We use the Taylor series expansion for small θ :

$$\sin \theta = \theta - \frac{\theta^3}{6} + O(\theta^5), \quad \tan \theta = \theta + \frac{\theta^3}{3} + O(\theta^5).$$

Let $\theta_n = \frac{x}{2^n}$.

$$\sin \theta_n - \tan \theta_n = \left(\theta_n - \frac{\theta_n^3}{6} \right) - \left(\theta_n + \frac{\theta_n^3}{3} \right) + O(\theta_n^5) = -\frac{\theta_n^3}{2} + O(\theta_n^5).$$

So, for fixed x , $u_n(x)$ is equivalent to:

$$u_n(x) \sim \left| -\frac{\left(\frac{x}{2^n}\right)^3}{2} \right|^{\frac{1}{2}} = \left(\frac{|x|^3}{2 \cdot 2^{3n}} \right)^{\frac{1}{2}} = \frac{|x|^{3/2}}{\sqrt{2}} \left(\frac{1}{2^{3/2}} \right)^n = \frac{|x|^{3/2}}{\sqrt{2}} \frac{1}{2^{3n/2}}.$$

Since $\sum \frac{1}{2^{3n/2}}$ is a convergent geometric series (ratio $r = 1/2^{3/2} < 1$), by the limit comparison test, the series $\sum u_n(x)$ converges for all $x \in \mathbb{R}$. The domain of convergence is $D = \mathbb{R}$. —

4. This is a power series in $(x-3)^2$. We use the Ratio Test for absolute convergence. Let $a_n(x) = \frac{n!(x-3)^{2n}}{n^{n+1}}$. We consider $L(x) = \lim_{n \rightarrow +\infty} \left| \frac{a_{n+1}(x)}{a_n(x)} \right|$.

$$\begin{aligned} \left| \frac{a_{n+1}(x)}{a_n(x)} \right| &= \left| \frac{(n+1)!(x-3)^{2(n+1)}}{(n+1)^{n+2}} \cdot \frac{n^{n+1}}{n!(x-3)^{2n}} \right| \\ &= |x-3|^2 \frac{(n+1)!}{n!} \frac{n^{n+1}}{(n+1)^{n+2}} \\ &= |x-3|^2 \left(\left(1 + \frac{1}{n}\right)^n \left(1 + \frac{1}{n}\right) \right)^{-1} \end{aligned}$$

We know $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n = e$ and $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right) = 1$.

$$L(x) = |x-3|^2 \cdot (e \cdot 1)^{-1} = \frac{|x-3|^2}{e}.$$

The series converges absolutely if $L(x) < 1$, i.e., $\frac{|x-3|^2}{e} < 1$, or $|x-3|^2 < e$.

$$|x-3| < \sqrt{e} \quad \Rightarrow \quad 3 - \sqrt{e} < x < 3 + \sqrt{e}.$$

The series diverges if $L(x) > 1$. If $L(x) = 1$, i.e., $|x-3|^2 = e$, we have $|x-3|^2 \frac{n^{n+1}}{(n+1)^{n+1}} = e \left(\frac{n}{n+1}\right)^{n+1} \sim e \cdot e^{-1} = 1$. The series diverges at the endpoints. The domain of convergence is $D = (3 - \sqrt{e}, 3 + \sqrt{e})$.

Exercise 4:

Consider the following series of functions:

$$g_n(x) = \frac{\sin x}{x^2 + 1} + \frac{\sin 2x}{x^2 + 2^2} + \dots + \frac{\sin nx}{x^2 + n^2} + \dots$$

1. Determine the domain of convergence D .
2. Study the uniform convergence on D .

Correction

1. **Domain of Convergence D :** The series is $\sum_{n=1}^{\infty} u_n(x)$, where $u_n(x) = \frac{\sin nx}{x^2 + n^2}$. The functions are defined for all $x \in \mathbb{R}$. We use the ****Weierstrass M-test**** for absolute and uniform convergence. We seek a bound M_n such that $|u_n(x)| \leq M_n$ for all $x \in \mathbb{R}$ and $\sum M_n$ converges.

$$|u_n(x)| = \left| \frac{\sin nx}{x^2 + n^2} \right| \leq \frac{|\sin nx|}{x^2 + n^2}.$$

Since $|\sin nx| \leq 1$ and $x^2 \geq 0$, we have $x^2 + n^2 \geq n^2$.

$$|u_n(x)| \leq \frac{1}{n^2}.$$

Let $M_n = \frac{1}{n^2}$. The series $\sum_{n=1}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ is a convergent p-series ($p = 2 > 1$). By the Weierstrass M-test, the series converges absolutely and uniformly on $\mathbb{R}$. The domain of convergence is $D = \mathbb{R}$.

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2. **Uniform Convergence on $D = \mathbb{R}$:** As shown in part 1, since $|u_n(x)| \leq M_n = \frac{1}{n^2}$ and $\sum M_n$ converges, the series $\sum u_n(x)$ converges ****uniformly**** on $\mathbb{R}$ by the ****Weierstrass M-test****.

Exercise 5:

Consider the series of functions with general term:

$$f_n(x) = \frac{nx}{1 + n^2x^2} - \frac{nx + x}{1 + n^2x^2 + 2n^2x + x^2}$$

1. Determine the domain of convergence D .
2. Calculate the sum of the series.
3. What is the nature of convergence of this series on D ?
4. Determine the nature of convergence of this series on $I =]0, +\infty[$ and $I = [a, +\infty[$ with $a > 0$.

Correction

1. **Domain of Convergence D :** The general term can be simplified:

$$f_n(x) = \frac{nx}{1 + (nx)^2} - \frac{(n+1)x}{1 + (n+1)^2x^2}.$$

The denominator of the second term is $1 + n^2x^2 + 2n^2x + x^2$. This is not quite $1 + ((n+1)x)^2 = 1 + (n^2 + 2n + 1)x^2 = 1 + n^2x^2 + 2nx^2 + x^2$. Let's re-examine the second term's denominator:

$$1 + n^2x^2 + 2n^2x + x^2 = 1 + (n^2 + 1)x^2 + 2n^2x.$$

It seems there is a typo in the original prompt, and the intended term was likely:

$$g_n(x) = \frac{nx}{1 + (nx)^2} - \frac{(n+1)x}{1 + ((n+1)x)^2}.$$

Assuming the general term is a telescoping difference:

$$f_n(x) = u_n(x) - u_{n+1}(x), \quad \text{where } u_n(x) = \frac{nx}{1 + (nx)^2}.$$

If $x = 0$, $f_n(0) = 0 - 0 = 0$. The series converges to 0. If $x \neq 0$, $u_n(x) \sim \frac{nx}{n^2x^2} = \frac{1}{nx}$. Thus, $\lim_{n \rightarrow +\infty} u_n(x) = 0$. The series is telescoping with partial sum:

$$S_N(x) = \sum_{n=1}^N f_n(x) = u_1(x) - u_{N+1}(x).$$

The limit of the partial sum is:

$$S(x) = \lim_{N \rightarrow +\infty} S_N(x) = u_1(x) - \lim_{N \rightarrow +\infty} u_{N+1}(x) = u_1(x) - 0 = u_1(x).$$

The series converges for all $x \in \mathbb{R}$. The domain of convergence is $D = \mathbb{R}$.

2. **Sum of the series:** Based on the telescoping form, the sum of the series is:

$$S(x) = u_1(x) = \frac{1 \cdot x}{1 + (1 \cdot x)^2} = \frac{x}{1 + x^2}.$$

3. **Nature of Convergence on $D = \mathbb{R}$:** We study the uniform convergence by checking the remainder term:

$$R_N(x) = S(x) - S_N(x) = \lim_{M \rightarrow +\infty} u_{M+1}(x) - u_{N+1}(x) = 0 - u_{N+1}(x) = -u_{N+1}(x).$$

$$R_N(x) = -\frac{(N+1)x}{1 + ((N+1)x)^2}.$$

We compute the supremum of the remainder:

$$|R_N| = \sup_{x \in \mathbb{R}} |R_N(x)| = \sup_{x \in \mathbb{R}} \left| \frac{(N+1)x}{1 + ((N+1)x)^2} \right|.$$

Let $y = (N+1)x$. Then $x = \frac{y}{N+1}$.

$$\sup_{y \in \mathbb{R}} \left| \frac{y}{1 + y^2} \right|.$$

Let $h(y) = \frac{y}{1+y^2}$. We find the maximum using derivatives: $h'(y) = \frac{1+y^2-y(2y)}{(1+y^2)^2} = \frac{1-y^2}{(1+y^2)^2}$. The critical points are $y = \pm 1$.

$$h(1) = \frac{1}{1+1} = \frac{1}{2}, \quad h(-1) = \frac{-1}{1+1} = -\frac{1}{2}.$$

Thus, $\sup_{y \in \mathbb{R}} |h(y)| = \frac{1}{2}$.

$$\sup_{x \in \mathbb{R}} |R_N(x)| = \frac{1}{2}.$$

Since $\lim_{N \rightarrow +\infty} \sup_{x \in \mathbb{R}} |R_N(x)| = \frac{1}{2} \neq 0$, the convergence is **not uniform** on $D = \mathbb{R}$.

4. **Nature of Convergence on $I =]0, +\infty[$ and $I = [a, +\infty[$ with $a > 0$:**

On $I =]0, +\infty[$: The convergence is **not uniform** since $\sup_{x>0} |R_N(x)| = 1/2 \neq 0$.

On $I = [a, +\infty[$ with $a > 0$: We check the remainder term $|R_N(x)| = \left| \frac{(N+1)x}{1 + ((N+1)x)^2} \right|$ for $x \geq a$. Let $g(y) = \frac{y}{1+y^2}$. The maximum of $g(y)$ for $y \geq (N+1)a$ is $g((N+1)a)$ if $(N+1)a \geq 1$, and $1/2$ otherwise (if the maximum $y = 1$ is in the domain). For N large enough such that $(N+1)a \geq 1$, the function $g(y)$ is decreasing for $y \geq 1$. For $N \geq \frac{1}{a} - 1$, the minimum value of $y = (N+1)x$ is at $x = a$, $y_{\min} = (N+1)a \geq 1$. The supremum is at $x = a$:

$$\sup_{x \geq a} |R_N(x)| = |R_N(a)| = \frac{(N+1)a}{1 + ((N+1)a)^2}.$$

The limit is:

$$\lim_{N \rightarrow +\infty} \sup_{x \geq a} |R_N(x)| = \lim_{N \rightarrow +\infty} \frac{(N+1)a}{1 + (N+1)^2 a^2} = \lim_{N \rightarrow +\infty} \frac{Na}{N^2 a^2} = \lim_{N \rightarrow +\infty} \frac{1}{Na} = 0.$$

Therefore, the convergence is ****uniform**** on $I = [a, +\infty[$ with $a > 0$.

Exercise 6:

Consider the series of functions with general term: $f_n(x) = (\sin x)^\alpha (\cos x)^n$.

1. Determine the sum and domain of convergence of this series.
2. Show that we have uniform convergence on any compact $[a, b]$ included in $]0, \frac{\pi}{2}[$.

Correction

1. **Domain of Convergence and Sum:** The series is $\sum_{n=0}^{\infty} f_n(x)$ (assuming $n \geq 0$ for the geometric series, or $n \geq 1$ for convergence testing; we'll assume $n \geq 0$). The series is a geometric series in $\cos x$:

$$\sum_{n=0}^{\infty} (\sin x)^\alpha (\cos x)^n = (\sin x)^\alpha \sum_{n=0}^{\infty} (\cos x)^n.$$

A geometric series $\sum r^n$ converges if and only if $|r| < 1$, and its sum is $\frac{1}{1-r}$. Here, the ratio is $r = \cos x$. The series converges if $|\cos x| < 1$.

$$|\cos x| < 1 \iff \cos x \neq 1 \text{ and } \cos x \neq -1.$$

This means $x \neq k\pi$ for any $k \in \mathbb{Z}$.

The term $(\sin x)^\alpha$ must also be defined. If α is not an integer, we require $\sin x > 0$.

- If $\alpha \in \mathbb{Z}$: $\sin x$ can be negative, but this doesn't change the convergence domain $D = \mathbb{R} \setminus \{k\pi, k \in \mathbb{Z}\}$.
- If $\alpha \notin \mathbb{Z}$: We need $\sin x > 0$, so $x \in \bigcup_{k \in \mathbb{Z}} (2k\pi, (2k+1)\pi)$. This requires us to assume α is a positive integer to align with the rest of the problem (specifically $]0, \frac{\pi}{2}[$ in part 2).

Assuming α is a positive integer (or the problem context implicitly suggests $x \in]0, \pi/2[$):

$$D = \mathbb{R} \setminus \{k\pi \mid k \in \mathbb{Z}\}.$$

The sum of the series on D is:

$$S(x) = (\sin x)^\alpha \cdot \frac{1}{1 - \cos x}.$$

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2. **Uniform Convergence on $[a, b] \subset]0, \frac{\pi}{2}[$:** On the interval $[a, b] \subset]0, \frac{\pi}{2}[$, we have $0 < a \leq x \leq b < \frac{\pi}{2}$. For all $x \in [a, b]$, we have $\cos x > 0$. Also, the ratio of the geometric series is $r(x) = \cos x$. On this compact interval, the maximum value of the ratio is $\max_{x \in [a, b]} \cos x = \cos a$. The minimum value is $\min_{x \in [a, b]} \cos x = \cos b$. Let $r_{\max} = \cos a$. Since $a > 0$, $r_{\max} < 1$. For all $x \in [a, b]$, the terms are bounded by:

$$|f_n(x)| = |(\sin x)^\alpha (\cos x)^n| = (\sin x)^\alpha |\cos x|^n.$$

Since $0 < \sin a \leq \sin x \leq \sin b < 1$, we have $0 < (\sin x)^\alpha \leq (\sin b)^\alpha$. Since $|\cos x| \leq \cos a = r_{\max}$, we have:

$$|f_n(x)| \leq (\sin b)^\alpha (\cos a)^n.$$

Let $M_n = (\sin b)^\alpha (\cos a)^n$. The series $\sum M_n$ is a geometric series with ratio $r = \cos a$. Since $0 < a < \frac{\pi}{2}$, we have $0 < \cos a < 1$. Thus, the series $\sum M_n$ converges. By the ****Weierstrass M-test****, the series converges ****uniformly**** on the compact interval $[a, b]$.